

# Trusted AI through Ethico-Legal Governors?

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VDW Summer School: Digitization, Artificial Intelligence and Responsibility, Berlin, June 30, 2022

# HOW TO START SUCH A TALK?

## LOVE

Well, I felt in love with the field of AI about 30 years ago  
because of a provocative statement  
by Prof. Jörg Siekmann in his AI lecture course

# AI & LOVE

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WHERE LOVE FALLS ...

... until recently:

... in today's online dating:

**Destiny**

determined by **AI-algorithms**



Swan "Petra" falls hopelessly in love with a pedal boat (2006)



# AI & LOVE

Peter and Katharina  
in love with "Saugi"



Pete und Cathy  
in love with "Dolly"



# WHAT IS AI?

A huge market

**Warum Deutschland mehr in KI investieren sollte**

## **Künstliche Intelligenz ist für Wirtschaft folgenreicher als die Dampfmaschine**

Von **Mirjam Hecking**

07.09.2018, 12.07 Uhr

**manager magazin**

The steam engine of the 21st century?

# WHAT IS AI?

AUTOMATISIERUNG

## Millionen Arbeitnehmer müssen umschulen

VON JANNIK WAIDNER - AKTUALISIERT AM 19.02.2021 - 09:04

Frankfurter Allgemeine  
ZEITUNG ● FAZ.NET

Source of social tensions and problems

# WHAT IS AI?

Great help in solving  
global environmental problems?

## How Innovative AI Solutions Can Help Combat Global Warming

Oct 14, 2020, 10:00am EDT



**Asokan Ashok** Forbes Councils Member

**Forbes Technology Council** COUNCIL POST | Paid Program

Innovation

Artificial intelligence / Machine learning

## Training a single AI model can emit as much carbon as five cars in their lifetimes

Deep learning has a terrible carbon footprint.

by **Karen Hao**

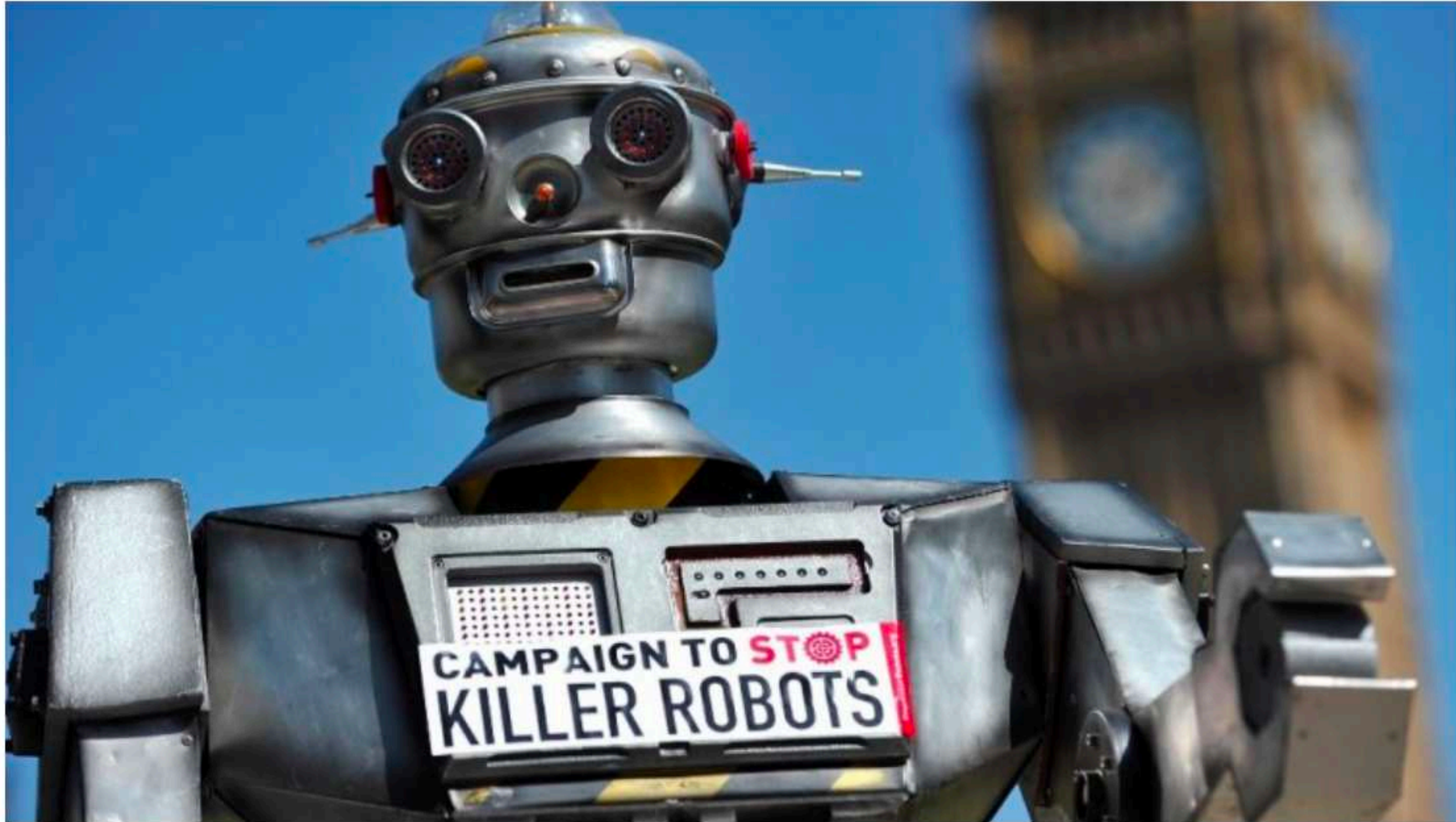
**MIT Technology Review**

June 6, 2019

Massive use of data-driven AI  
could itself become a climate killer?



# WHAT IS AI?



Source: Deutschlandfunk/AFP, Carl Court, 2019

A danger regarding military tension and escalation

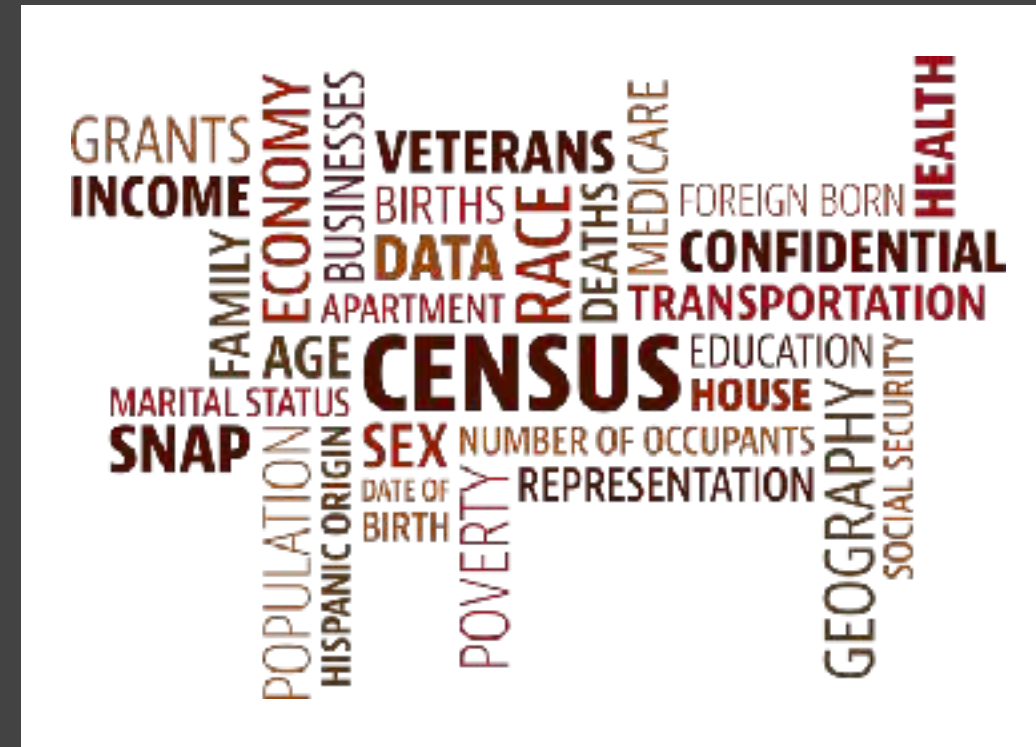


# WHAT IS AI?

## Machine Learning (ML) Hype

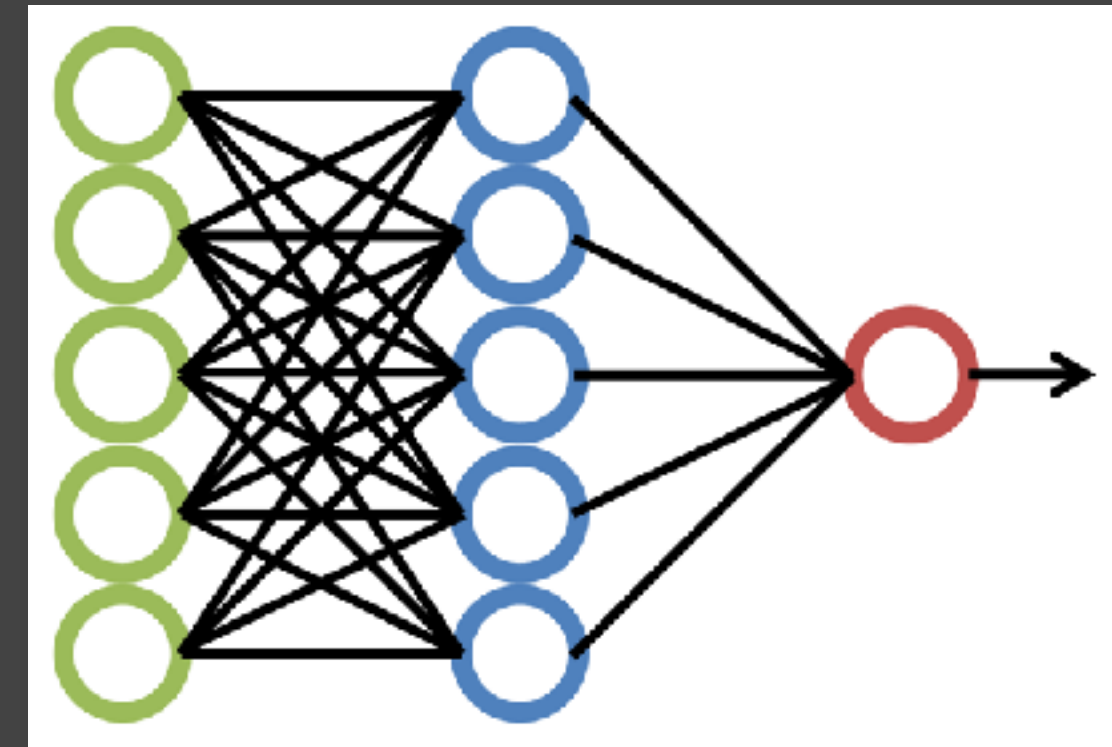
Many currently believe:

AI =



Data

+



ML/Deep Learning

Aha ... really?



# WHAT IS AI?

Elena

Carla

can compute  
has humor  
has emotions  
has language  
everything  
humans can  
a priori  
competencies?





# WHAT IS AI?

Weak AI: ... solve specific problems

Strong AI: ... everything humans can do (and possibly way more)

Strong AI requires at least (own working hypothesis):

1. Problem Solving (in the sense of weak AI)
2. Exploring and Acting in Unknown Territory (e.g. Mars-Rover)
3. Abstract & Rational Reasoning (e.g. exploration of a new theory in maths)
4. Self-Reflection (e.g. detect own mistakes, questioning the results of one's own thinking)
5. Social Interaction (e.g. adjusting personal goals and values to those of a community)

Machine Learning

Hybrid AI



# AI-FRIEND FOR THE SMART CITY

We want to raise and train him in real life

What hides  
in the couch?

A companion  
for life?

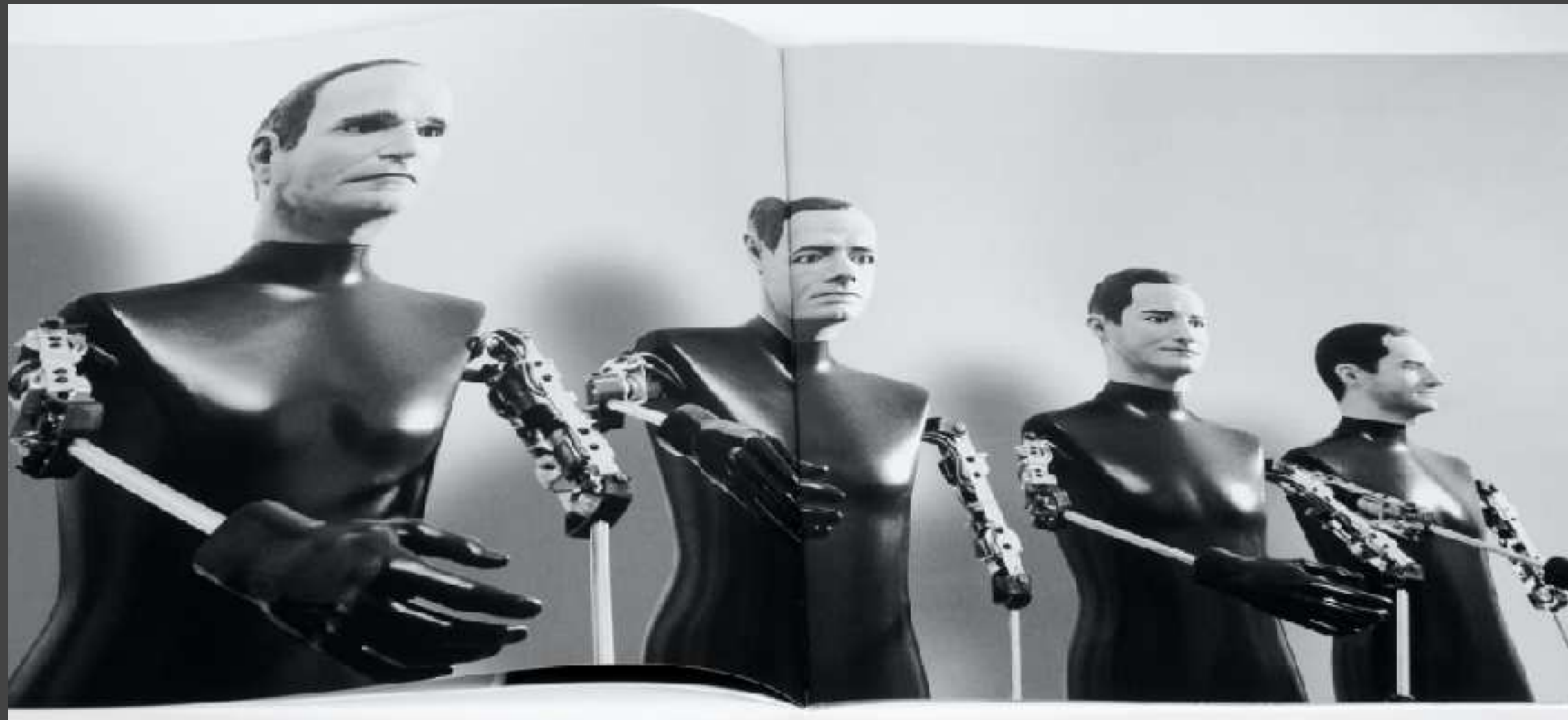
Hmm, maybe  
we better  
recheck  
with Neill?





# WHAT IS AI?

Fascinating research  
"Man vs. Machine"



Societal challenge  
"Responsible, trusted AI systems"



# WHAT IS AI?

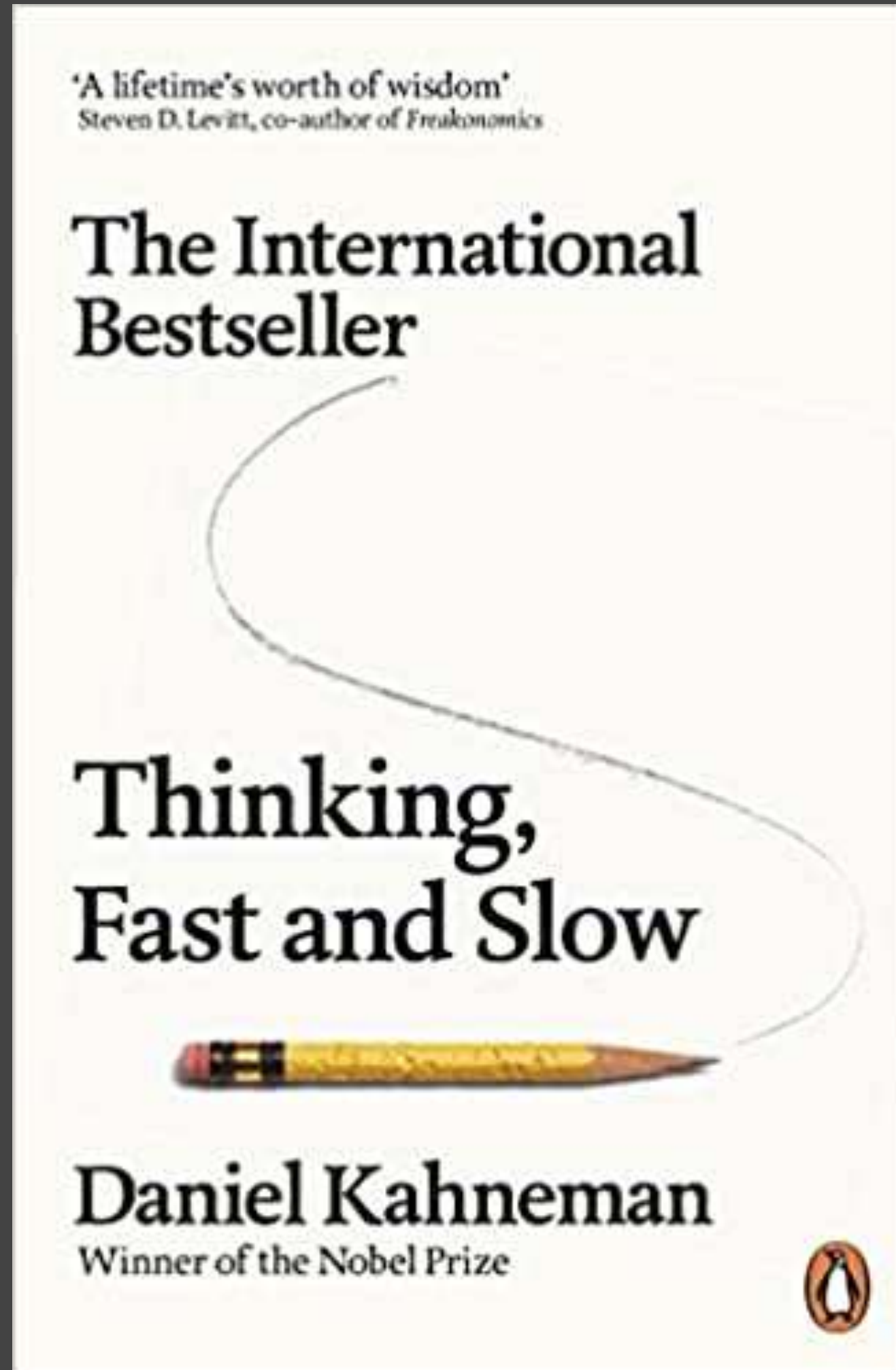


- Do our current AI systems know what they are doing?
- Do we know what we are doing when we leave increasingly critical decisions to such systems?
- Is normative directionlessness and unpredictability a core character of future AI systems?

I argue for: Normative, societal control for AI (in critical applications!)



# HOW DOES MAN DO IT?



System 2

(Analogy: Rule-based AI)

SLOW RATIONALE  
REFLECTION & CONTROL

HYBRID AI



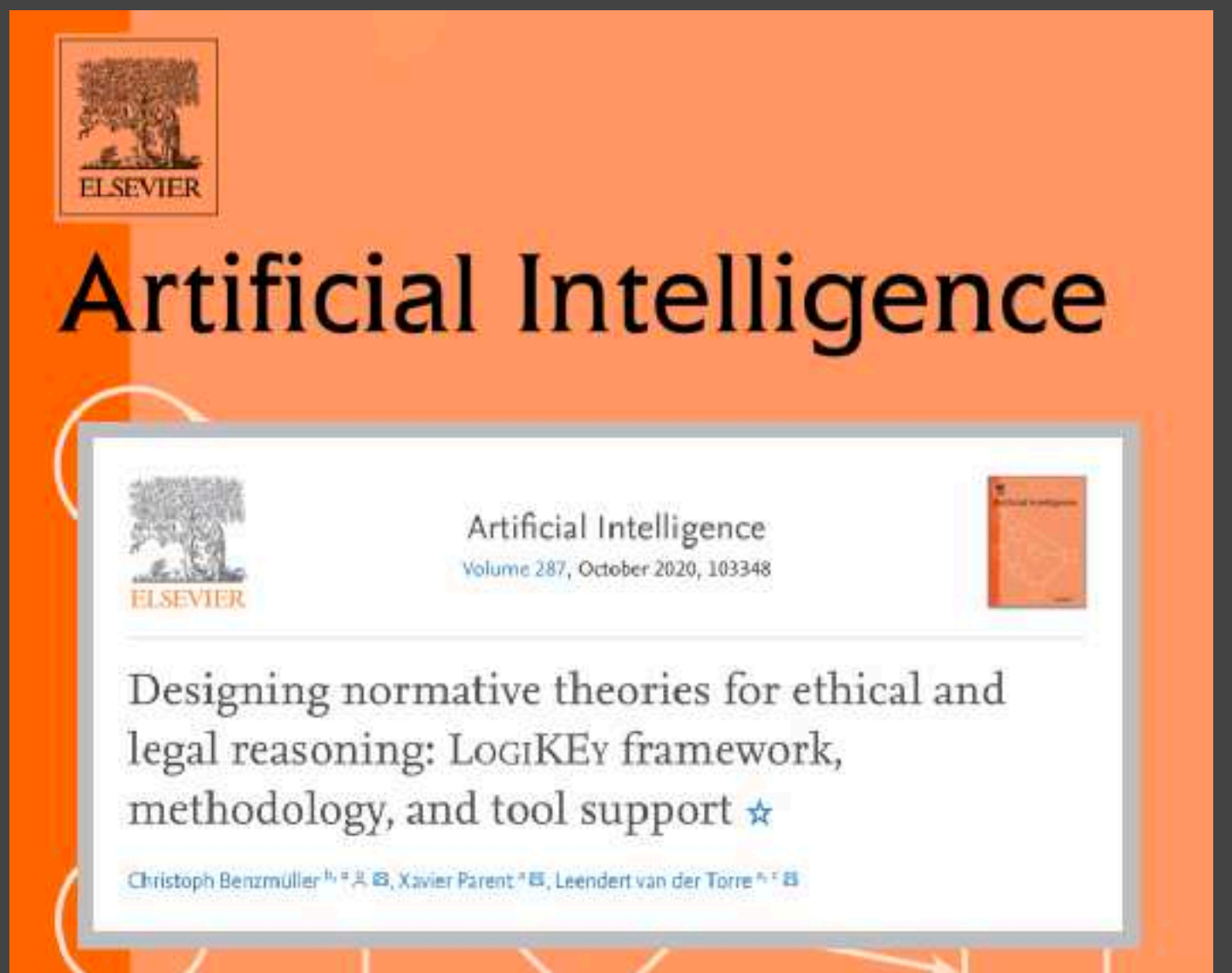
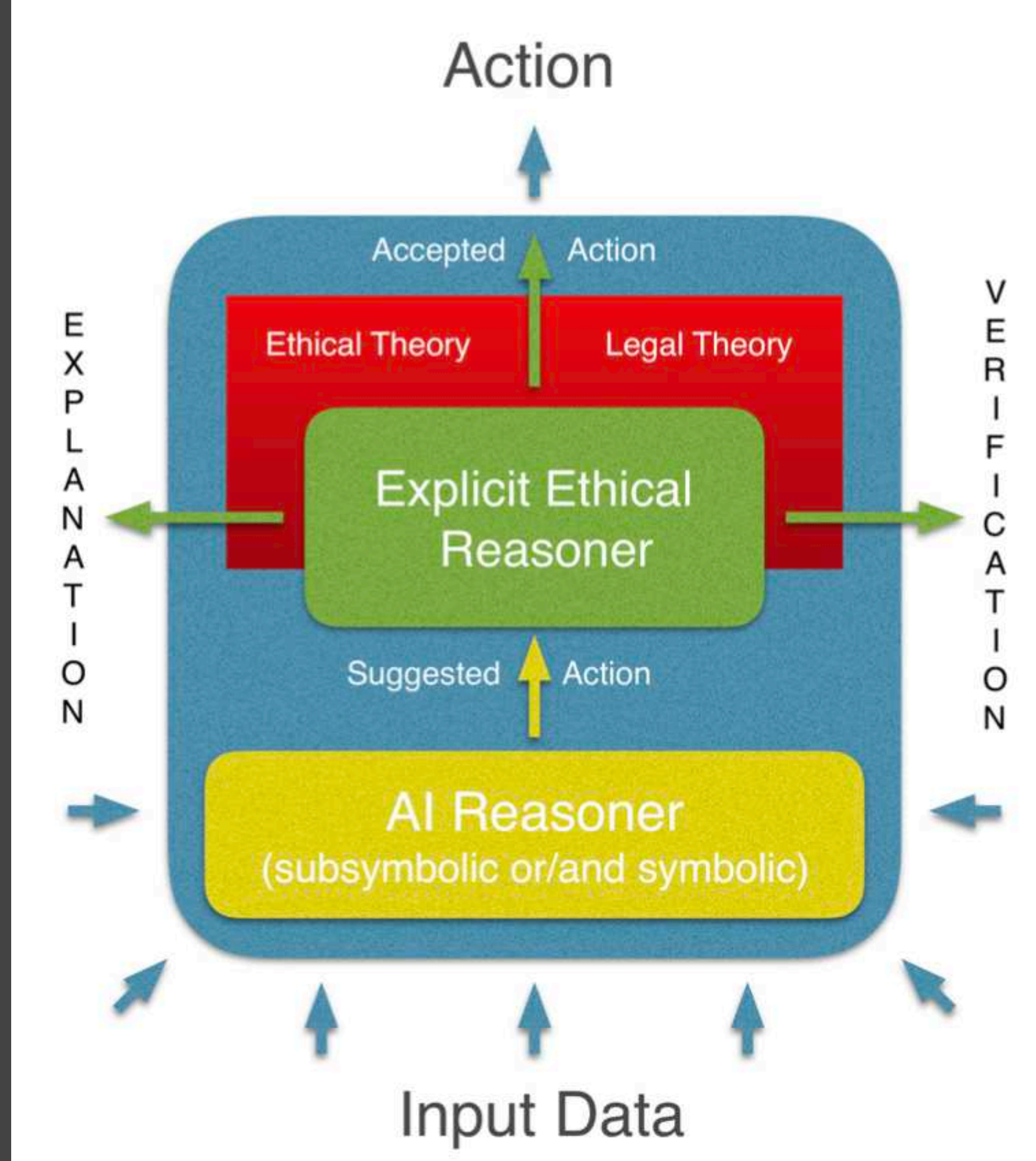
System 1

FAST INTUITIVE ACTION

(Analogy: Data-driven AI)



# ETHICO-LEGAL GOVERNORS?





# RAISING AN AI?



Today: AI on the leash



Old fashioned: dog on leash





# RAISING AN AI?

The girls want their AI-companion to understand the environment



... and for this they sometimes scold and set limits



# NOTE THE CRUCIAL ROLE OF LANGUAGE!



Thou shalt not  
kill, steal, ... !

Language & rational argumentation:

- important control function
- communication of norms/limits
- highly efficient & effective
- ...
- (indispensable for strong AI)

A **simulation** of language understanding is not sufficient for this purpose!

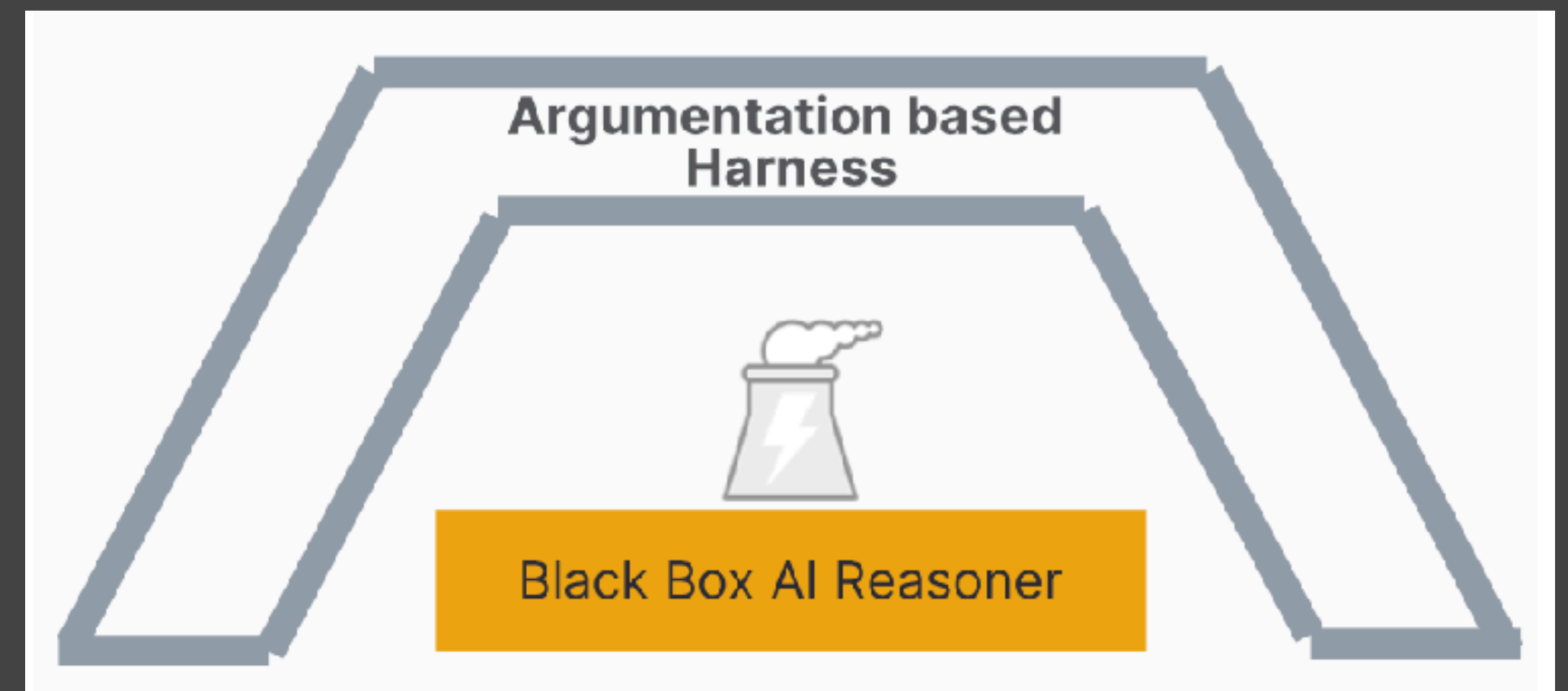
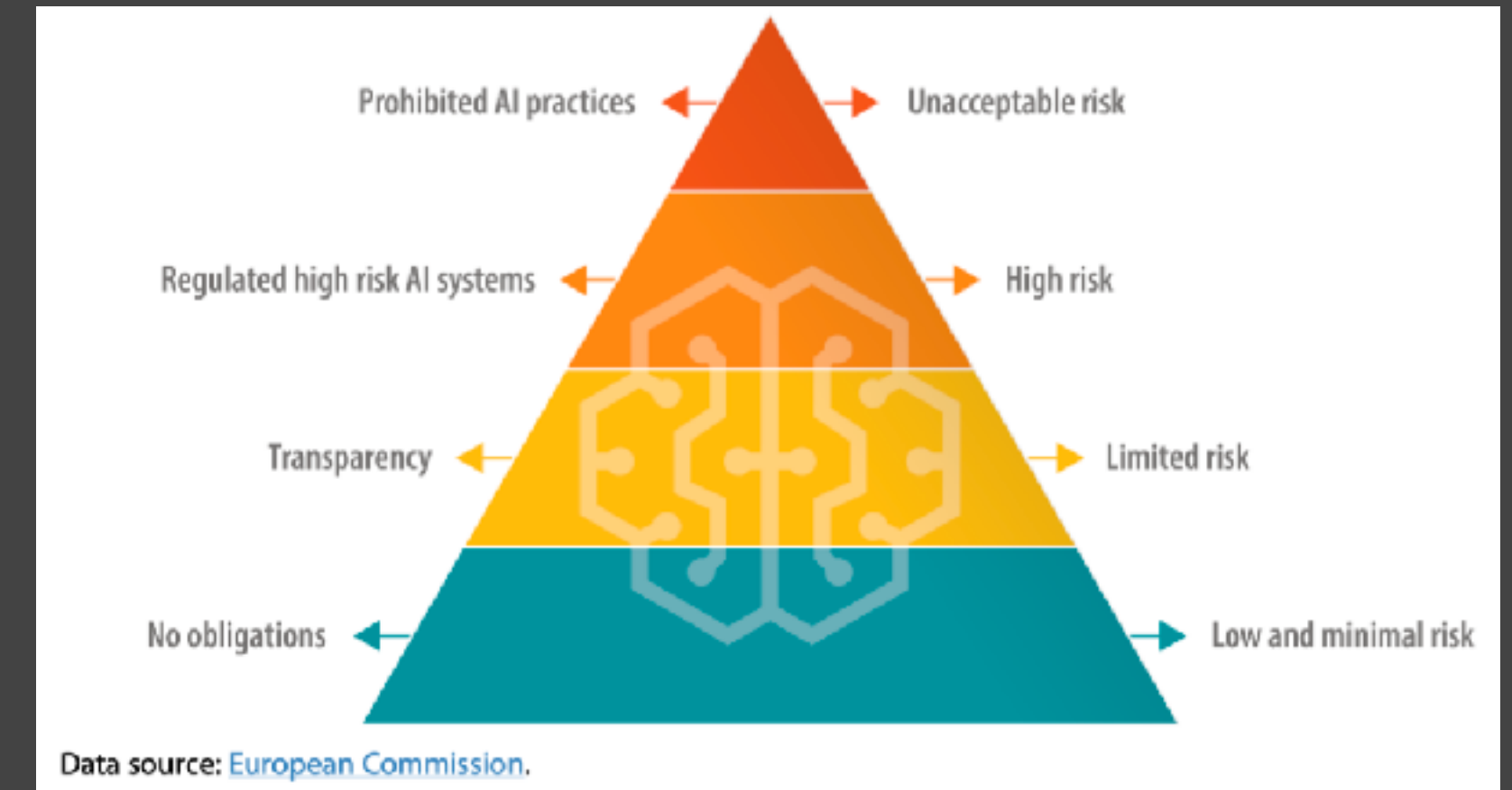


# CONCLUSION

High risk AI applications: →  
How to control? How to create trust?

- Opening AI black box systems?
  - Not a good idea (for applications of high criticality)!
  - Limited impact on trust
  - Invitation for adversarial attacks

I argue for control harnesses  
resembling the **System 2** layer  
as mentioned earlier





# Some further Reading

(preprints: <http://christoph-benzmueller.de/publications.html>)

## LogiKEy Methodology & Universal (Meta-)Logical Reasoning

- \_\_, Parent & van der Torre (2020), **Designing normative theories for ethical and legal reasoning: LogiKEy framework, methodology, and tool support**. Artificial Intelligence. (<https://doi.org/10.1016/j.artint.2020.103348>)
- Benzmüller et.al. (2020), **LogiKEy Workbench: Deontic Logics, Logic Combinations and Expressive Ethical and Legal Reasoning (Isabelle/HOL Dataset)**. Data in Brief (<https://doi.org/10.1016/j.dib.2020.106409>)
- Benzmüller (2019), **Universal (meta-)logical reasoning: Recent successes**. Sci. Comp. Progr. (<http://doi.org/10.1016/j.scico.2018.10.008>)

## Foundational Studies in Ethics & Law

- Fuenmayor & Benzmüller (2018), **Formalisation and Evaluation of Alan Gewirth's Proof for the Principle of Generic Consistency in Isabelle/HOL**. (<https://www.isa-afp.org/entries/GewirthPGCProof.html>)
- \_\_ (2019), **Harnessing Higher-Order (Meta-)Logic to Represent and Reason with Complex Ethical Theories**. PRICAI 2019 ([https://doi.org/10.1007/978-3-030-29908-8\\_34](https://doi.org/10.1007/978-3-030-29908-8_34))
- Benzmüller & Fuenmayor (2021), **Value-oriented Legal Argumentation in Isabelle/HOL**. ITP 2021 (<https://doi.org/10.4230/LIPIcs.ITP.2021.7>)
- \_\_, Fuenmayor & Lomfeld (2022), **Modelling Value-oriented Legal Reasoning in LogiKEy**. (<https://arxiv.org/abs/2006.12789>)